

A/B Test Analysis

E-Commerce Checkout Page Test

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Analysis Summary

This document contains the complete A/B test analysis for the e-commerce checkout page experiment, including data exploration, cleaning, statistical analysis, and visualization.

1. Initial Data Exploration

Prompt: What files can you see in this folder right now?

Working Directory: /home/user/claude

Files Found: test_brief.txt (216 bytes)

2. Data File Overview

Prompt: Read ecommerce_ab_test.csv and describe the data structure

Column Name	Description
Visitor_ID	Unique identifier for each visitor
Assignment_Date	Date visitor was assigned to test
Test_Variant	Which variant (Control/A/B/Treatment)
Landing_Page	control_hero_v1 or new_hero_v2
Device_Category	Mobile, Desktop, or Tablet
Traffic_Source	Source of traffic (Organic, Social, Paid, Email, etc.)
Country	Geographic location
Visitor_Status	New or Returning
Session_Minutes	Duration of session
Pages_Viewed	Number of pages visited
Added_To_Cart	Whether added items to cart
Completed_Purchase	KEY METRIC: Whether purchase completed
Order_Value_USD	Revenue metric: Purchase amount

3. Data Quality Assessment

Prompt: Analyze distinct values and data completeness

Test_Variant Values (Inconsistent Formatting):

Value	Count
A	2074
B	2067
Variant B	741
Control	729
b	596
a	564
treatment	362
CONTROL	341
A (with space)	294
B (with space)	292

Completed_Purchase Values (Inconsistent Formatting):

Value	Count
No	5460
no	1162
N	756
Yes	502
yes	107
Y	73

Data Completeness Issues:

Issue	Count	Impact
Duplicate Visitor_IDs	60	7% of data
Blank Session_Minutes	570	7.1% of data
Blank Traffic_Source	418	5.2% of data

4. Data Cleaning Process

Prompt: Collapse variants and purchase values, deduplicate visitors, save as ab_test_clean.csv

Cleaning Steps Applied:

1. **Test_Variant Collapse:** Mapped all variants to two groups: A (Control, A, a, ' A') and B (B, b, Variant B, treatment, ' B')
2. **Completed_Purchase Standardization:** Converted all values to lowercase: Yes/Y/yes → 'yes', No/N/no → 'no'
3. **Deduplication:** Kept only the first occurrence of each Visitor_ID, removing 60 duplicate records
4. **Final Dataset:** 8,000 unique visitors (3,976 in Group A, 4,024 in Group B)

5. A/B Test Results

Prompt: Calculate conversion rates and statistical summary

Group	Visitors	Purchases	Conversion Rate
A (Control)	3,976	281	7.07%
B (Variant)	4,024	397	9.87%
Difference	—	+116	+2.80pp (39.6%)

Key Metrics:

- **Absolute Improvement:** 2.80 extra purchases per 100 visitors
- **Relative Improvement:** 39.6% higher conversion rate for Group B
- **Additional Revenue:** 116 more purchases from the new design variant

6. Executive Summary

We tested two checkout page designs with 8,000 visitors split evenly. The control design (Group A) converted 281 of 3,976 visitors at 7.07%, while the new design (Group B) converted 397 of 4,024 visitors at 9.87%. That's a 2.80 percentage point improvement, or 39.6% higher conversion for the variant.

This test doesn't tell us whether the improvement will hold with different traffic sources, seasonal changes, or after visitors get familiar with the new design.

Data Limitations:

- Inconsistent data formatting required standardization
- 60 duplicate Visitor_IDs were removed
- 570 sessions missing duration data (7.1%)
- 418 sessions missing traffic source (5.2%)
- No statistical significance testing performed
- No analysis of segment-specific performance (device, traffic source, geography)

7. Analysis Deliverables

Files Created:

1. **test_brief.txt** - Brief description of landing pages and conversion definition
2. **ab_test_clean.csv** - Cleaned dataset with standardized variants, deduplicated visitors, 8,000 rows
3. **ab_result.html** - Interactive bar chart visualization with conversion rates and visitor counts